

Does AI Help or Hurt Learning? *

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Abstract

AI is transforming how students learn. This development raises pressing concerns about whether it promotes equitable access to education—or widens existing learning gaps. Who will benefit from AI in their studies, and who may be left behind? We examine this question through a pre-registered lab experiment ($N = 572$) with two stages: a study phase, where participants learn a novel topic, followed by an exam on the same topic. Participants were randomly assigned to one of three conditions that determined what tools they could use while studying: (1) Control (browsing only), (2) AI-assisted (access to a generative AI), or (3) AI-guided (AI access with additional guidance). During the study phase, students could allocate their time between reading materials, solving practice questions, and using the assigned tools. All participants then completed an exam without AI access, allowing us to causally estimate the effects of AI on learning. First, we find no overall effect of AI access on learning outcomes. However, this average masks significant heterogeneity: high-GPA women clearly benefit from AI, while men show mixed responses. Second, we find that students with AI access attempt fewer practice questions, meaning that students with access to AI engage less in active problem solving.

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Finally, we analyze participants' prompt data to explore how students interact with AI and how this affects learning. Together, the results show that AI reshapes not only who benefits from learning, but how learning itself takes place.

1 Introduction

The rapid advancement of generative AI is set to transform several spheres of society. Among these, education stands out, as AI tools are poised to drastically change how students acquire knowledge. Through AI, a student today can access PhD-level writing and analysis with a simple prompt, rather than reaching this level of understanding through their own effort over time. This raises a fundamental question about the technology’s impact on human capital accumulation: *does AI access help or hurt learning?*

The impact of AI on student learning remains unclear and likely depends on *how* students use these tools. AI may enhance learning by complementing student effort—or undermine it by substituting active engagement. Its effects may therefore differ across student groups, raising concerns about widening educational gaps. To the best of our knowledge, we provide the first lab study on AI and learning. This approach allows us to isolate causal effects of AI on a number of outcomes such as study behavior, and distributional effects. We identify who benefits, who is left behind, and what can be done to mitigate emerging gaps.

We design and pre-register an experiment to answer these questions. Despite high interest in the topic due to the ongoing debate on whether and how to integrate generative AI into education, empirical evidence on its impact on learning remains scarce (Yan et al., 2024). The main challenge is that AI use is unobservable, making it difficult for field studies to restrict control groups from AI use. Few studies have aimed to establish causal effects (Bastani et al., 2024; Lehmann et al., 2024), and have been criticized due to features such as teacher quality that are hard to control in field settings (Tan and Rajaratnam, 2024). Our controlled lab experiment allows us to clearly identify the effects of AI on learning by ensuring full control over AI access—something that is difficult to achieve in field settings. This approach offers several important advantages. We can precisely control access to AI tools, ensuring a clear differentiation between treatment and control groups and avoiding spillovers. Our homogeneous setting guarantees that all participants learn about the same topic in the same way. Furthermore, our design enables a detailed analysis of distributional impacts across different student subpopulations, and our unique prompting data allows us to identify specific mechanisms through which AI use may help or hurt students. While

the laboratory setting may be less realistic than the field, we design incentives that mirror real-course conditions and teach the material in a way that reflects how university students learn in practice. The experiment consists of three between subject treatment variations. Students in the lab are asked to go over a lecture and work on practice questions to learn about a topic that is new to them (Esperanto).¹ In the baseline treatment, students learn without access to AI but with access to Google Search, the status quo before the emergence of generative AI tools. In the AI-access treatment, students learn with access to AI, but with no formal guidance on how to use this tool except for some general instructions. In the AI-guided treatment, students learn with access to AI and are given guidance on how to use it best to promote their own learning.

Our experiment yields three key findings on the impact of AI tools on learning outcomes. First, contrary to widespread concerns among educators, we find no significant negative effects of AI use on learning, as measured by both exam performance and students' self-assessed learning. In fact we find that AI access *neither* hurts nor helps students on average. Second, our results reveal that this average zero effect masks significant heterogeneous effects across student subpopulations. While high-GPA women clearly benefit from AI-guided learning, male students show mixed responses. In particular, high-GPA men are less likely to score at the top or at the bottom, while low-GPA men experience increased variance in performance. Put differently, top women benefit from AI in their learning, while men -on average- remain unaffected. This means that AI risks exacerbating the pre-existing gender gap in learning outcomes as women -who have dominated men in educational outcomes for decades² - may end up doing even better. Third, we find that providing AI access changes the way students allocate their study time between chatting with the bot and active problem solving. Students in both AI treatment groups attempt significantly fewer practice questions within the same time limit as those in the control group who only have access to browsing. The reduced number of questions attempted when AI is available suggests a shift in attention away from active problem-solving toward chatting above and beyond the time spent

¹Esperanto is a constructed international language designed for easy learning and communication, with an estimated 100,000 to 2 million speakers worldwide. Its unfamiliarity to our sample and its simplicity make it ideal for our experiment, allowing students to learn a significant amount in a short time.

²See for instance [Voyer and Voyer \(2014\)](#)

browsing in the control treatment. This shift in time use may hinder skill development in the longer run.

Taken together, our findings suggest that AI can change not only educational outcomes but also how students learn in the first place. Women with High-GPA are expected to benefit the most from the AI-supported study, despite being the group with the lowest adoption rates in previous research (Carvajal et al., 2024). In contrast, mixed effects for male students raise concerns that AI could amplify existing educational inequalities. This is particularly relevant in light of the documented 'boy crisis' in education in OECD countries (Autor and Wasserman, 2013; Autor et al., 2023). Moreover, if students increasingly substitute AI interaction for active problem-solving, this change in study behavior may hinder the development of core skills and worsen learning outcomes over time.

2 Experimental Design

We conducted the experiment in the CedEX Lab at the University of Nottingham in December 2024. Participants (N=572) are current undergraduate and graduate students from programs in all fields of study across the university.

The experiment was designed to mimic the stages of a typical university course, where students first learn about the new topic, then deepen and practice the content through practice questions, and finally are tested through an exam. To test how generative AI affects learning, we experimentally varied students' access to AI tools during the second stage of the experiment (completion of practice questions). In particular, participants were randomly assigned to one of three treatments, which are explained in more details below: T1: no access to ChatGPT (Control), T2: access to ChatGPT (AI-assisted), and T3: guided access to ChatGPT (AI-guided). All participants, independently of treatment status, learned about the same topic: Esperanto – a constructed international language designed for easy learning and communication. Its simplicity and unfamiliarity to the sample made it ideal for measuring how well students learn new material, rather than relying on their prior knowledge.³

³This is a key design choice in our study because previous knowledge of a topic may discourage students to use AI since they can answer the questions themselves. We would then not be able to estimate the effect of

Each session adhered to the same procedure:

1. **Stage 1:** Students learn Esperanto for 15 minutes using written learning materials provided by the researchers.
2. **Stage 2:** Students complete practice questions with access to different learning aids depending on treatment for about 20 minutes. This is the stage where the treatment variations take place.
3. **Stage 3:** Students take an exam testing their learning without access to any notes or learning aids.
4. **Stage 4:** Students take a post-study survey asking them about demographics as well as their learning experience during the session and attitudes and beliefs about AI.⁴

The entire experiment was conducted in Qualtrics. The learning materials and practice questions were presented across different pages, which allowed participants to move back and forth. More questions than possible to answer were included to accommodate different learning speeds. When the time for the lesson and practice questions was up, the screen moved automatically to a wait screen so that everybody would start the exam at the same time. There was no time limit on the exam but it was calibrated to last about 10 minutes on average. The total duration of each session was about one hour.

2.1 T1: No access to ChatGPT

T1 was designed to mirror learning before the advent of generative AI and to work as a baseline against which we can measure the impact of (guided) AI access on learning outcomes. As students had access to a web browser and online learning before the advancement of generative AI, we provided participants in this group free access to a web browser during stage 2. Websites providing access to AI chatbot sites, however, were blocked in T1. See instructions for students in Figure ??.

AI on learning.

⁴Basic questions such as gender, experience with AI and languages spoken are asked at the beginning of the study, before stage 1.

2.2 T2: Access to ChatGPT

T2 was designed to mirror learning when ChatGPT is available to students without clear guidelines, structure and training—a situation in which it is up to each student whether and how she or he wants to use AI in the learning process. This mirrors the situation which many universities face as they are grappling with the important task of adjusting their policies to the advent of AI.

Students accessed ChatGPT on their assigned computer browser with a already logged in premium ChatGPT account. The account was provided by the researchers and was cleared of any memory. The prompting history was archived and saved by researchers for analysis of prompting. See instructions for students in Figure ??.

2.3 T3: Access to Guided ChatGPT

This treatment is the exact same as T2, except that students now also receive guidance (written by the researchers) on how to use ChatGPT. The idea behind this treatment is to explore the potential of providing not just equal access but also equal training and encouragement in AI use for students. See instructions for students in Figure ??.

2.4 Incentives

We aimed to mimic the incentives provided in typical college courses in the lab. The incentives in most real courses around the world are to assign a smaller percentage of the final grade to assignments and practice questions and to have a larger weight on a final exam or midterms and a final exam. The idea of the assignments with lower percentages is to incentivize students to cover the materials and prepare them for the exam(s). In the same spirit, we offered incentives to reward correct responses and help students deepen their learning in the Esperanto lesson, while offering higher incentives for every correct answer in the exam. The structure of incentives is as follows:

- GBP 5 for completing the study
- GBP 7 for correctly answering at least 20 out of over 50 practice questions

- GBP 1 per correct answer in the test (maximum amount GBP 15)

The maximum payment they could earn is GBP 27.

From the beginning of the session students were informed that the most important part of the session (and where they can earn the most) is the exam. This is emphasized throughout the session. The intention is that they focus their efforts of trying to learn as much as possible from all the resources they have available, just as they would in a real-life course.

3 Empirical Strategy

3.1 Outcomes of Interest

We focus on three primary measures of learning. First, our main variable of interest is exam performance, captured through both an objective and a subjective metric. The objective measure is the total number of correct answers on the final exam (ranging from 0 to 15). The subjective measure is students' self-assessed learning, recorded as an indicator equal to one if they slightly or strongly agreed with the statement "I feel like I learned a lot from solving practice questions" (on a four-point scale), and zero otherwise. For both measures, we investigate heterogeneous effects by background characteristics such as gender and GPA.

Second, we examine distributional outcomes by estimating the likelihood of students scoring in the top or bottom of the test score distribution. A top scorer is prespecified as having more than 10 correct answers, while a low scorer is defined as solving less than 5.

Finally, we assess students' exam preparedness by analyzing two measures: the number of practice questions solved correctly and the amount of time spent completing the final exam.

3.2 Econometric Specifications

The main specification regresses the outcomes presented above on indicators for the two AI treatments, keeping the control as the excluded indicator:

$$y_i = \alpha_0 + \alpha_1 \text{AI-assisted}_i + \alpha_2 \text{AI-guided}_i + \varepsilon_i \quad (1)$$

The main specification adds baseline covariates to control for different levels of AI adoption and students characteristics.⁵

We assess whether there are any gender differences in responses to the treatment using the specification:

$$y_i = \beta_0 + \beta_1 \text{AI-assisted}_i + \beta_2 \text{AI-guided}_i + \beta_3 \text{Female}_i + \beta_4 \text{AI-assisted}_i \times \text{Female}_i + \beta_5 \text{AI-guided}_i \times \text{Female}_i + \varepsilon_i \quad (2)$$

Where the coefficients β_1 and β_2 measure the effects of the treatments among men, β_3 measures any gender gaps in the control group, and β_4 and β_5 measure the differential effects of the treatments for women.

A similar specification will be presented replacing the Female variable with High GPA, a measure of academic skill equal to 1 if students are in first-class honours (GPA 70% or above) and 0 otherwise.

A final specification involves the interaction effects between treatments, gender and GPA. Following the previous literature, this specification may allow us to see differential effects by gender and GPA simultaneously. For example, if the treatments affect high-GPA female students differentially.

$$y_i = \gamma_0 + \gamma_1 \text{AI-assisted}_i + \gamma_2 \text{AI-guided}_i + \gamma_3 \text{Female}_i + \gamma_4 \text{High GPA}_i + \gamma_5 \text{AI-assisted}_i \times \text{Female}_i + \gamma_6 \text{AI-assisted}_i \times \text{High GPA}_i + \gamma_7 \text{AI-guided}_i \times \text{Female}_i + \gamma_8 \text{AI-guided}_i \times \text{High GPA}_i + \gamma_9 \text{AI-assisted}_i \times \text{Female}_i \times \text{High GPA}_i + \gamma_{10} \text{AI-guided}_i \times \text{Female}_i \times \text{High GPA}_i + \varepsilon_i \quad (3)$$

The triple interactions will measure whether top female students are differentially af-

⁵The covariates include: Age 21 or younger, undergraduate student, indicators for field of study, use AI occasionally or all the time, has a paid subscription to an AI provider. In regressions without interactions with gender and high GPA, we add these variables as covariates.

affected by the treatments. The double interactions with High GPA will measure if top male students are differentially affected by the treatments.

The results from these specifications are presented in table form in the appendix.

3.3 Balance across Treatments and Descriptive Statistics

Table 1 reports baseline characteristics by treatment status. Overall, the randomization produced groups that are similar along most observed dimensions. The shares of women, younger students, undergraduates, field of study, prior AI use, and paid AI subscriptions are close across the control, AI-assisted, and AI-guided groups. One variable is not balanced: the share of high-GPA students⁶ is lower in the AI-guided group than in the control group (22.6% versus 36.8%). In all our results, we control for baseline covariates.

The experimental sample consists of 604 students. In terms of composition, about 61% of participants are women, 57% are aged 21 or younger, and 64% are undergraduate students. Academic background is mixed across fields of study, and about 34% of participants are classified as high-GPA students. Prior exposure to AI tools is common: roughly 63% report using AI occasionally or all the time, while 7% report having a paid AI subscription. Because 22 students left the survey while working on the exam, we do not have outcomes for them. We further drop 10 students who do not report a binary gender. Thus, the analytical sample consists of 572 students.

Figures 1 and 2 provide a non-parametric view of the exam performance outcome distributions. Figure 1 shows substantial overlap in exam-score distributions across the three treatment arms in the full sample, consistent with the absence of large average treatment effects. Figure 2 suggests, however, that the distributions differ somewhat across subgroups. In particular, the score distribution for women and for high-GPA students appears somewhat more right-shifted under AI access, whereas the distributions for men and low-GPA students show less favorable patterns relative to the control group. The left tails of the distributions are typically longer in the control group, in particular for men (Figure 2b) and high-GPA students (Figure 2c). In the distribution for men, it also appears that not many achieve top

⁶In the UK, there is no numeric GPA, so we take as high GPA the students who expect to get a first-class honours degree (GPA of 70% or above).

scores in the control group, but more men are able to achieve high scores under the AI treatments (Figure 2b). The distribution of high-GPA students under the AI-guided treatment seems more compressed (i.e., not many obtaining low scores or high scores) relative to both the control and AI-assisted distributions (Figure 2c).

When we look at average exam performance, we do not see any differences as students in each treatment condition obtain a score of 8 (out of 15) on average (Figure 3). However, the descriptive differences in the density plots speak to the fact that the average effects hides substantial heterogeneity they anticipate the heterogeneous treatment effects that we will discuss in the next section.

4 Results

4.1 No Overall Impact of AI on Learning

We begin by examining whether access to generative AI affects learning. This is a central outcome in the policy debate over whether AI tools should be restricted or incorporated into educational settings, given concerns that they may reduce students' own learning.

Table 2 reports the effects of the two AI treatments on our two learning outcomes: exam performance in a setting without access to external aids, and students' self-assessed learning. Across specifications, we find no evidence of an overall effect of AI access on learning. In columns (1) and (5), assignment to the two AI treatments have small effects and none of the estimates is statistically distinguishable from zero. Relative to a control-group mean of around 8 correct answers in exam performance and between 81% and 83% of students indicating that they somewhat agree or agree with the statement "I feel like I learned a lot working through the questions.", these effects are economically small.

Columns (2), (3), (6), and (7) of Table 2 report the pre-registered heterogeneity analyses by gender and GPA. These specifications do not show broad, systematic differences in the effects of AI on exam performance or self-assessed learning when heterogeneity is examined along only one dimension at a time. Taken together, they support the conclusion that AI access has no average effect on learning and provide little evidence of heterogeneity by

gender or GPA separately.

Columns (4) and (8) estimate a fully saturated model with treatment effects allowed to vary jointly by gender and GPA. While the triple interactions are not statistically significant, their magnitudes are economically meaningful. Most notably, the interaction between AI-guided access, female, and high GPA is associated with 1.4 additional correct answers on the exam, about 17% of the relevant control-group mean. For self-assessed learning, the corresponding triple interaction is -19.9 pp. This estimate should be interpreted relative to the positive coefficient for high-GPA men in the same specification (20.6 pp), implying a reversal in sign: while high-GPA men report higher perceived learning under AI-guided access, high-GPA women report lower perceived learning. Although we are underpowered to estimate these triple interactions precisely, the pattern suggests that high-GPA women may benefit disproportionately from AI-guided access in terms of measured performance while perceiving that they learn less, in sharp contrast to high-GPA men.

Figure 4 complements these regression results by plotting subgroup-specific treatment coefficients for exam performance. These subgroup coefficients are useful for visualizing how the estimates vary across cells defined jointly by gender and GPA, but they were not separately pre-registered; our pre-analysis plan pre-specified the interaction analyses reported in the tables. The figure reinforces the main result of no overall treatment effect, while also suggesting that heterogeneity may be concentrated in more specific subgroups rather than along gender or GPA alone. In particular, point estimates under AI-guided access are more favorable for high-GPA women, but more negative for low-GPA women and high-GPA men (Figure 4b). Under AI-assisted access (Figure 4a), the corresponding subgroup patterns are smaller and less systematic. The confidence intervals are wide because we are not powered enough to identify small effects in the subsamples, so these subgroup differences should be interpreted cautiously.

We find a similar absence of average effects for students' self-assessed learning. Figure 5 shows that point estimates are generally close to zero and imprecisely estimated across subgroups in both AI treatments. Taken together, the evidence in Table 2 and Figure 4 indicates that providing AI access neither improves nor worsens learning on average in our

setting, even though some subgroup-specific patterns emerge in the estimates.

4.2 Heterogeneous Effects Emerge in the Tails of the Score Distribution

The absence of average effects on exam performance masks meaningful heterogeneity in where students end up in the score distribution. To examine this more directly, Table 3 reports treatment effects on two pre-registered binary outcomes: obtaining a top score (more than 10 correct answers) and obtaining a low score (fewer than 5 correct answers). Figure 6 complements these estimates by plotting subgroup-specific treatment coefficients for the same outcomes.

At the aggregate level, neither treatment changes the probability of reaching the top or the bottom of the score distribution. In column (1), the effects on obtaining a top score are close to zero for both AI-assisted and AI-guided. Likewise, in column (5), the effects on obtaining a low score are economically small and statistically insignificant in the full sample. Thus, the null average effects in test scores are not driven by uniform shifts in performance across all students.

The heterogeneity results point instead to differential effects across student groups, particularly under AI-guided access. The clearest pattern appears for low scores. In column (7), AI-guided increases the probability of scoring in the bottom tail by 4.6 percentage points among the baseline group, while the interaction with High GPA is negative and statistically significant (-12.3 pp), implying that the adverse effect is concentrated among lower-GPA students. A similar, though less precisely estimated, pattern appears for AI-assisted access. These results suggest that AI access can increase the risk of very poor performance among academically weaker students, even if average scores remain unchanged.

For top scores, the evidence is less clear. The interaction specifications in columns (2)–(4) do not indicate broad average gains in the probability of reaching the top tail, but they do suggest meaningful differences once gender and GPA are considered jointly. In particular, the fully interacted specification for AI-guided access in column (4) implies a larger representation among top scorers for high-GPA women and a lower representation for high-GPA men. Figure 6 makes this pattern even more clear: under AI guidance, the point estimate

for obtaining a top score is most positive for high-GPA women, whereas the estimates are weaker or negative for low-GPA women and high-GPA men. At the same time, the bottom-tail results in the figure show that low-GPA students and, in particular, low-GPA men are the group most likely to experience an increase in very poor performance under both AI treatments.

Taken together, Table 3 and Figure 6 suggest that AI does not shift the entire score distribution upward or downward. Instead, its effects are concentrated in the tails and differ sharply across subgroups. In our setting, AI-guided access appears to help some students reach high performance while increasing the risk that other students, especially lower-GPA students, end up at the bottom of the distribution.

4.3 AI Changes Study Behavior: Fewer Practice Questions Attempted

A natural question is whether access to AI changes how students allocate study time. On the one hand, AI may substitute for active engagement with traditional course material if students rely on the chatbot for answers rather than working through practice problems themselves. On the other hand, AI could complement conventional study tools by helping students ask follow-up questions or explore material in greater depth. Understanding how AI affects study behavior is therefore important, particularly if time spent with AI crowds out traditional practice.

Table 4 and Figure 7 examine two measures of study behavior. The first outcome is time spent on the final exam, a setting in which no student had access to AI and no time limit was imposed. The second is the number of practice questions attempted during the study session, which took place under the assigned treatment condition and within a fixed 37-minute window.⁷ Because time is fixed during the study period, the number of attempted practice questions provides a useful measure of how students allocate their effort across AI use and conventional practice.

Columns (1)–(4) of Table 4 show no meaningful effect of either treatment on time spent on the final exam. Across all specifications, the estimated effects of AI-assisted and AI-guided

⁷The 37 minutes were split into about 15 minutes to study an Esperanto lesson and the remaining time for working on practice questions.

access are small and statistically insignificant relative to an average of 10.8 minutes for women and 10.4 minutes for men in the control group. This is unsurprising, since AI was not available during the exam itself.

By contrast, columns (5)–(8) show that AI access substantially reduces the number of practice questions attempted during the study session. In the control group, students attempt between 33.8 and 35.6 practice questions on average. Assignment to the AI-assisted treatment lowers this by 4.17 questions, while assignment to the AI-guided treatment lowers it by 5.77 questions, both highly statistically significant. These correspond to reductions of about 12 and 17%, respectively for each treatment, relative to the control-group mean. Thus, while AI does not change how long students spend on the final test, it clearly changes how they study beforehand, leading students to attempt fewer conventional practice questions.

The pre-registered heterogeneity specifications in columns (6) and (7) do not show broad, systematic differences by gender or GPA considered separately. Figure 7 is nevertheless informative about how these effects vary across narrower subgroups. The figure suggests that the decline in attempted practice questions is concentrated among men, especially high-GPA men, whereas high-GPA women appear much less affected. In the fully interacted specification in column (8), the estimated reduction under AI-guided access is about 5.2 questions for the baseline group, but the positive triple interaction for high-GPA women is economically large, implying a much smaller decline for that subgroup. These subgroup differences are not estimated precisely enough to support strong claims, but they are consistent with the broader pattern in the paper that high-GPA women appear to use AI in a way that is less substitutive of traditional study effort.

Importantly, attempting fewer questions does not necessarily imply lower efficiency. Appendix Table A1 shows that students in both AI treatment groups solve a larger share of attempted practice questions correctly, with little evidence of subgroup differences. The results therefore suggest a shift in study strategy: students with AI attempt fewer questions, but do so with a higher success rate. That is, studying with AI takes longer but does not meaningfully affect performance in the final exam where no AI tools were allowed.

Taken together, the results suggest that high-GPA women are the subgroup that benefits

most from AI-guided access. Unlike most other students, they show only a modest decline in attempted practice questions relative to their control counterparts and are simultaneously more represented among top scorers, which is consistent with a less substitutive and potentially more effective use of AI during study.

5 Who Benefits from AI? Descriptive Evidence from Prompting Behavior

6 Discussion

Our findings provide insights on the (heterogeneous) effects of AI on learning. The clear winners appear to be high-GPA women who perform better than men and than students in the control group in the AI guided treatment. Our evidence shows that a key reason for why this is the case is that they attempt as many practice questions as control students despite the overall and strong negative effect on number of attempted questions in the AI treatments. This suggests that they may have a more efficient use of AI. They self report that they use AI not just to answer the questions and that AI made them feel engaged with the materials. Paradoxically, high-ability women is the group that has been identified as having the lower adoption rates of AI ([Carvajal et al., 2024](#)). The analysis of the ChatGPT prompts will allow us to gain more insights as to why high-GPA women disproportionately benefit from AI use.

The potential losers in the AI revolution are the male and low GPA students. Not only were men attempting fewer practice questions but their performance was also below the control group. This can be seen more clearly in Figure ??, which plots the kernel densities of the exam scores for the whole sample and the main subgroups. The density for men in the AI treatments is well below and to the left of the control group. Students with low GPAs have a similar picture, while it is the opposite (AI has higher mass and to the right than the control distribution) for high-GPA students.

Based on these findings, it seems plausible that AI can exacerbate existing inequalities in education that have been summarized by the boy crisis, a term that was coined to char-

acterize the phenomenon that boys are falling behind in education with detrimental consequences on their later-life outcomes (Autor and Wasserman, 2013; Autor et al., 2023). For example, in most OECD countries, a larger proportion of boys than girls do not attain the baseline level of proficiency in any of the core subjects, namely mathematics, reading and science, and drop out at higher rates than girls (OECD, 2015).

Overall, we see that some students may benefit from using AI in the learning process, in particular when there is guidance on how to use these tools. Our next steps will focus on analyzing the prompts of students in the AI treatments. This exercise will provide deeper insights into behavioral differences and help uncover why AI benefits some students more than others.

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7 Tables

Table 1: Treatment balance

	(1) Control (T1)	(2) AI-assisted (T2)	(3) AI-guided (T3)	(4) Diff. (T1)-(T2)	(5) Diff. (T1)-(T3)
Female	0.618 (0.487)	0.613 (0.488)	0.602 (0.491)	0.004 (0.049)	0.016 (0.049)
21 years old or younger	0.584 (0.494)	0.520 (0.501)	0.613 (0.488)	0.063 (0.049)	-0.029 (0.049)
Undergraduate student	0.641 (0.481)	0.602 (0.491)	0.683 (0.466)	0.039 (0.048)	-0.042 (0.047)
Arts	0.124 (0.331)	0.112 (0.316)	0.116 (0.321)	0.012 (0.032)	0.009 (0.032)
Economics or Business School	0.201 (0.402)	0.224 (0.418)	0.241 (0.429)	-0.024 (0.041)	-0.040 (0.041)
Engineering	0.163 (0.370)	0.163 (0.371)	0.131 (0.338)	-0.001 (0.037)	0.032 (0.035)
Medicine and Health Sciences	0.163 (0.370)	0.153 (0.361)	0.146 (0.354)	0.010 (0.036)	0.017 (0.036)
Social Sciences	0.153 (0.361)	0.199 (0.400)	0.176 (0.382)	-0.046 (0.038)	-0.023 (0.037)
Other	0.196 (0.398)	0.148 (0.356)	0.191 (0.394)	0.048 (0.037)	0.005 (0.039)
High GPA	0.368 (0.484)	0.429 (0.496)	0.226 (0.419)	-0.060 (0.049)	0.142*** (0.045)
Use AI occasionally or all the time	0.627 (0.485)	0.612 (0.488)	0.648 (0.479)	0.015 (0.048)	-0.021 (0.048)
Paid AI subscription	0.057 (0.233)	0.056 (0.231)	0.090 (0.288)	0.001 (0.023)	-0.033 (0.026)
Observations	209	196	199	405	408

Notes. This table reports means and standard deviations (in parentheses) for the covariates listed in the left-hand column for the control group (Column 1), the AI-assisted treatment (Column 2), and the AI-guided treatment (Column 3). Columns 4 and 5 report the mean differences between each treatment group and the control group, with standard errors in parentheses. The treatment was assigned within each laboratory section by designating a third of the available computers to each treatment condition and randomly allocating participants to computers. Statistical significance of the differences in Columns 4 and 5 is indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Treatment effects on exam performance and self-assessed learning

	Test score				Self-assessment of learning			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI-assisted	0.078 (0.192)	-0.224 (0.295)	-0.046 (0.236)	-0.464 (0.383)	0.004 (0.040)	-0.049 (0.066)	-0.014 (0.052)	-0.038 (0.088)
AI-guided	-0.234 (0.189)	-0.211 (0.293)	-0.318 (0.217)	-0.066 (0.332)	-0.041 (0.041)	-0.069 (0.067)	-0.068 (0.048)	-0.119 (0.083)
Female		-0.047 (0.262)		0.110 (0.287)		-0.025 (0.056)		-0.001 (0.069)
AI-assisted × Female		0.498 (0.386)		0.667 (0.485)		0.087 (0.083)		0.039 (0.108)
AI-guided × Female		-0.039 (0.390)		-0.403 (0.440)		0.045 (0.085)		0.081 (0.102)
High GPA			0.006 (0.290)	0.288 (0.441)			-0.049 (0.059)	-0.012 (0.093)
AI-assisted × High GPA			0.310 (0.397)	0.471 (0.616)			0.047 (0.083)	-0.021 (0.135)
AI-guided × High GPA			0.258 (0.437)	-0.554 (0.651)			0.095 (0.094)	0.206 (0.135)
Female × High GPA				-0.452 (0.593)				-0.059 (0.120)
AI-assisted × Female × High GPA				-0.269 (0.815)				0.116 (0.170)
AI-guided × Female × High GPA				1.362 (0.878)				-0.199 (0.187)
Constant	7.448*** (0.311)	7.546*** (0.322)	7.514*** (0.311)	7.539*** (0.318)	0.828*** (0.067)	0.854*** (0.069)	0.843*** (0.068)	0.857*** (0.075)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean women	7.968	7.968	7.968	7.968	0.808	0.808	0.808	0.808
Mean men	8.184	8.184	8.184	8.184	0.829	0.829	0.829	0.829
Observations	572	572	572	572	572	572	572	572

Notes. This table reports estimates from Equation 3 (pre-registered) for two primary outcomes. Exam score is measured on a 0–15 scale. Self-assessed learning is an indicator equal to one if the respondent selected “somewhat agree” or “agree” to the statement, “I feel like I learned a lot working through the questions,” and zero otherwise (“disagree” or “somewhat disagree”). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Treatment effects on scoring at the top or bottom

	Top score				Low score			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI-assisted	0.009 (0.028)	0.016 (0.041)	0.034 (0.035)	0.057 (0.048)	-0.003 (0.020)	0.040 (0.029)	0.029 (0.024)	0.073* (0.043)
AI-guided	-0.002 (0.026)	0.026 (0.038)	0.010 (0.029)	0.078* (0.043)	0.009 (0.020)	0.040 (0.030)	0.046** (0.023)	0.073** (0.036)
Female		0.043 (0.035)		0.065* (0.038)		0.029 (0.025)		0.015 (0.020)
AI-assisted × Female		-0.011 (0.056)		-0.038 (0.068)		-0.070* (0.039)		-0.071 (0.051)
AI-guided × Female		-0.047 (0.053)		-0.109* (0.058)		-0.050 (0.043)		-0.044 (0.048)
High GPA			0.057 (0.042)	0.097 (0.062)			0.056* (0.032)	0.031 (0.035)
AI-assisted × High GPA			-0.061 (0.059)	-0.104 (0.090)			-0.081* (0.043)	-0.078 (0.063)
AI-guided × High GPA			-0.036 (0.062)	-0.175** (0.070)			-0.123*** (0.039)	-0.117** (0.052)
Female × High GPA				-0.065 (0.083)				0.038 (0.060)
AI-assisted × Female × High GPA				0.069 (0.121)				-0.006 (0.086)
AI-guided × Female × High GPA				0.236** (0.116)				-0.007 (0.076)
Constant	0.022 (0.042)	0.012 (0.042)	0.010 (0.043)	-0.015 (0.042)	0.103*** (0.038)	0.079** (0.034)	0.081** (0.036)	0.064** (0.031)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean women	0.088	0.088	0.088	0.088	0.056	0.056	0.056	0.056
Mean men	0.053	0.053	0.053	0.053	0.013	0.013	0.013	0.013
Observations	572	572	572	572	572	572	572	572

Notes. This table reports estimates from the pre-registered specification in Equation 3 for two binary outcomes. The thresholds for these outcomes were pre-specified as follows: Top scorer equals one if the participant answered more than 10 questions correctly, and low scorer equals one if the participant answered fewer than 5 questions correctly. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

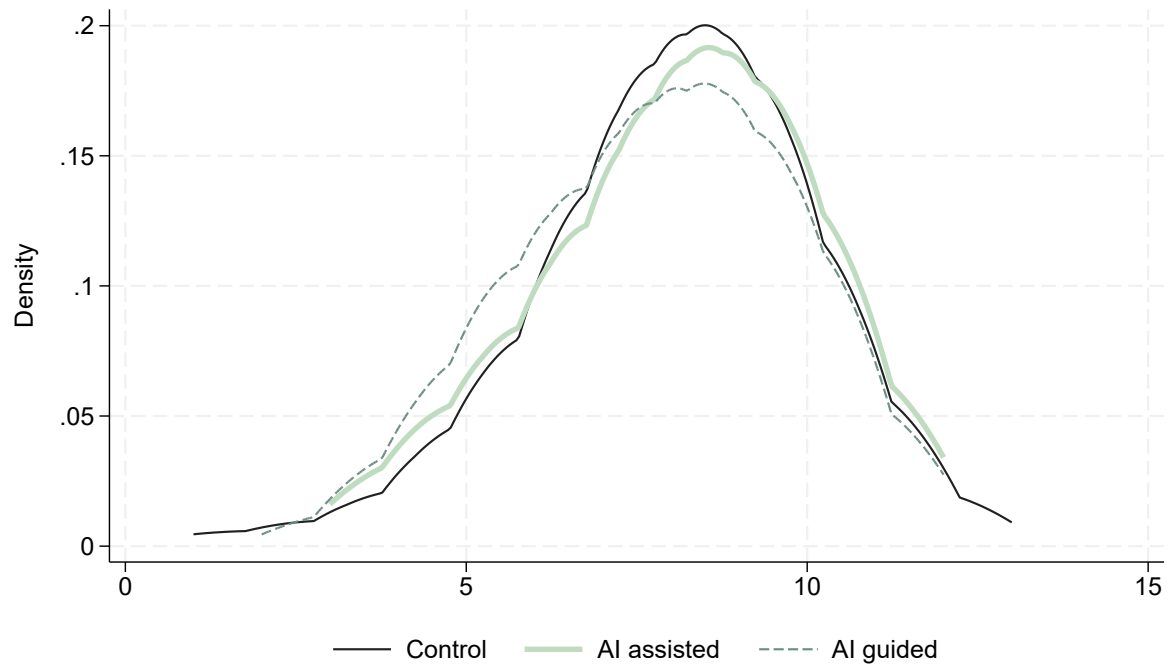
Table 4: Treatment effects on number of practice questions attempted

	Time spent on test (minutes)				Number of practice questions attempted			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI-assisted	-0.132 (0.248)	0.162 (0.420)	-0.056 (0.315)	0.346 (0.550)	-4.171*** (1.094)	-6.255*** (1.710)	-4.337*** (1.351)	-5.022** (2.121)
AI-guided	0.149 (0.243)	0.225 (0.371)	0.090 (0.290)	0.426 (0.441)	-5.773*** (1.125)	-6.412*** (1.751)	-6.166*** (1.305)	-5.243*** (1.852)
Female		0.479 (0.377)		0.904** (0.455)		-1.865 (1.538)		0.085 (1.691)
AI-assisted × Female		-0.484 (0.518)		-0.649 (0.668)		3.379 (2.227)		1.037 (2.760)
AI-guided × Female		-0.124 (0.494)		-0.540 (0.583)		1.048 (2.333)		-1.469 (2.580)
High GPA			0.279 (0.385)	1.004* (0.609)			1.107 (1.695)	4.415 (2.884)
AI-assisted × High GPA			-0.174 (0.518)	-0.560 (0.865)			0.460 (2.320)	-3.237 (3.724)
AI-guided × High GPA			0.266 (0.541)	-0.435 (0.815)			1.396 (2.567)	-3.327 (3.998)
Female × High GPA				-1.169 (0.783)				-5.357 (3.484)
AI-assisted × Female × High GPA				0.555 (1.072)				6.197 (4.731)
AI-guided × Female × High GPA				1.129 (1.092)				7.767 (5.175)
Constant	10.112*** (0.435)	9.989*** (0.472)	10.109*** (0.446)	9.758*** (0.510)	30.538*** (1.991)	31.447*** (2.025)	30.718*** (1.946)	30.525*** (2.072)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean women	10.819	10.819	10.819	10.819	33.857	33.857	33.857	33.857
Mean men	10.447	10.447	10.447	10.447	35.628	35.628	35.628	35.628
Observations	572	572	572	572	594	594	594	594

Notes. This table presents results from the pre-registered specification in Equation 3 for two outcomes. Time spent is measured as the total minutes spent on the exam. Practice questions solved is the total number of practice questions attempted. There were 90 practice questions in total. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

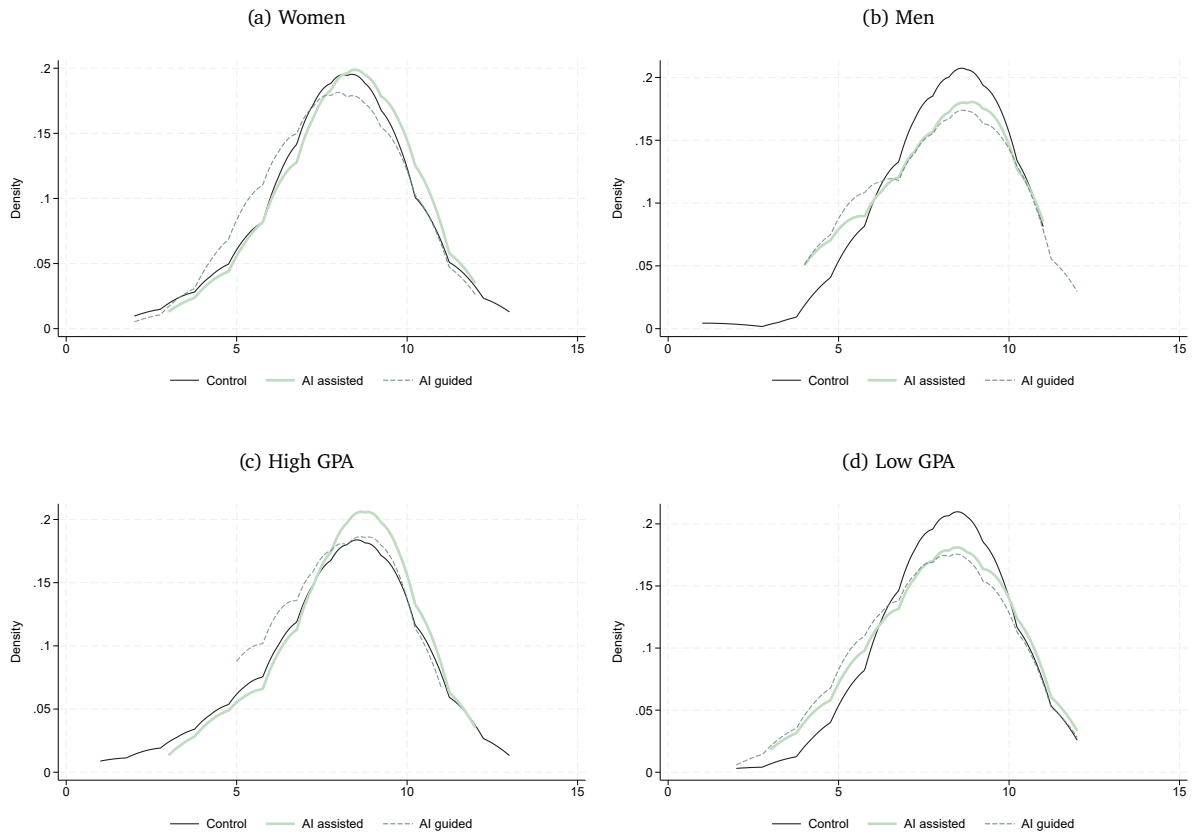
8 Figures

Figure 1: Density of exam performance



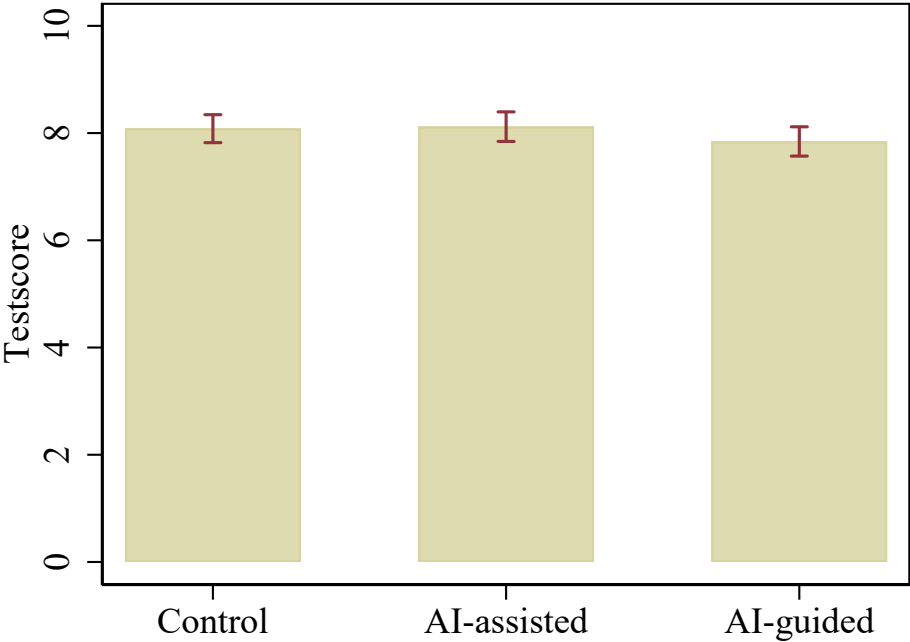
Notes. This figure presents kernel density estimates using the Epanechnikov kernel function with the width of the density window around each point equal to 1, separately for each treatment arm.

Figure 2: Density of exam performance by gender and GPA



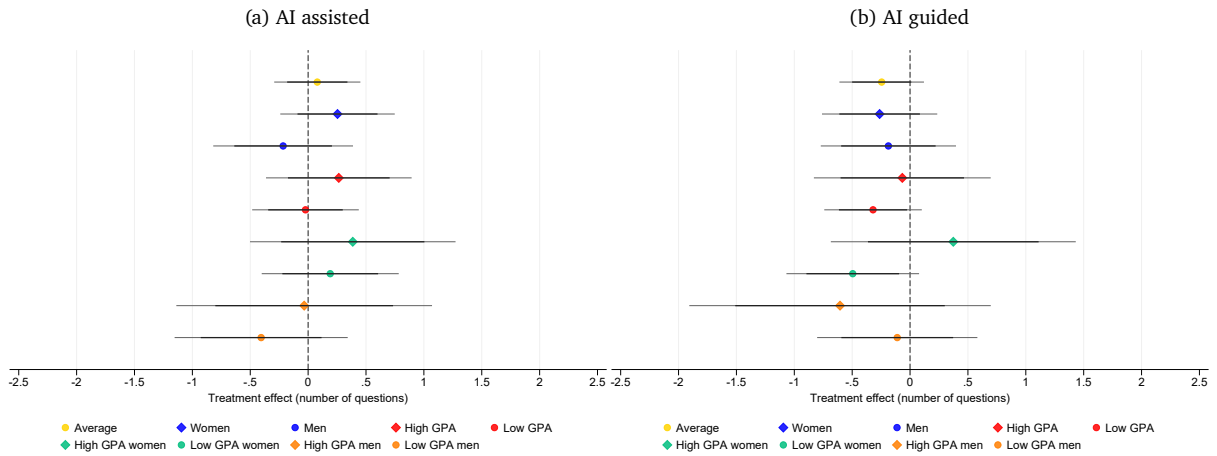
Notes. These figure presents kernel density estimates using the Epanechnikov kernel function with the width of the density window around each point equal to 1, separately for each treatment arm. The panels define the sample split on which the estimates are generated: gender and GPA.

Figure 3: Exam performance by treatment



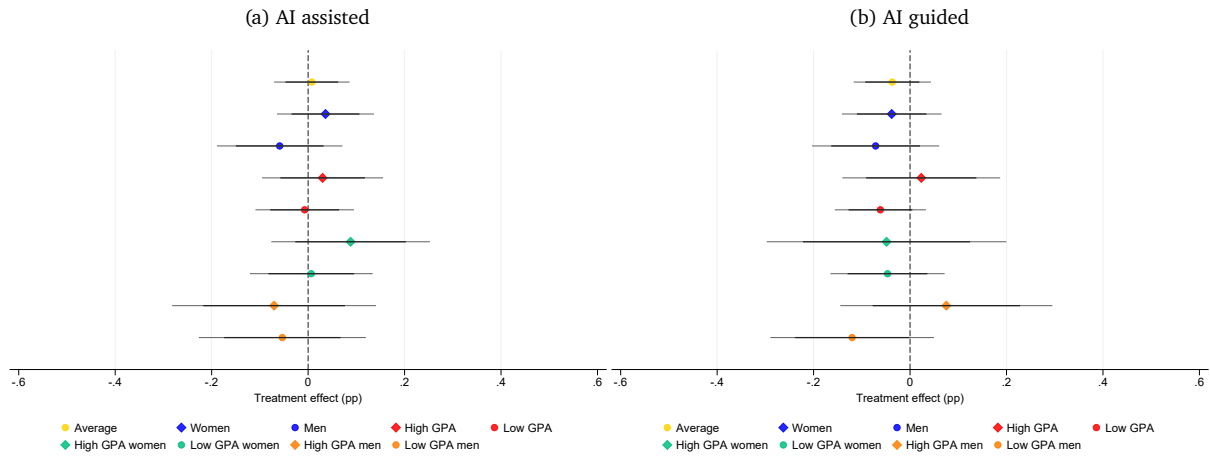
Notes. Each bar plots the raw mean score in the Esperanto exam for students each treatment condition, along with 95% confidence bars.

Figure 4: Treatment effects on exam performance by subgroups



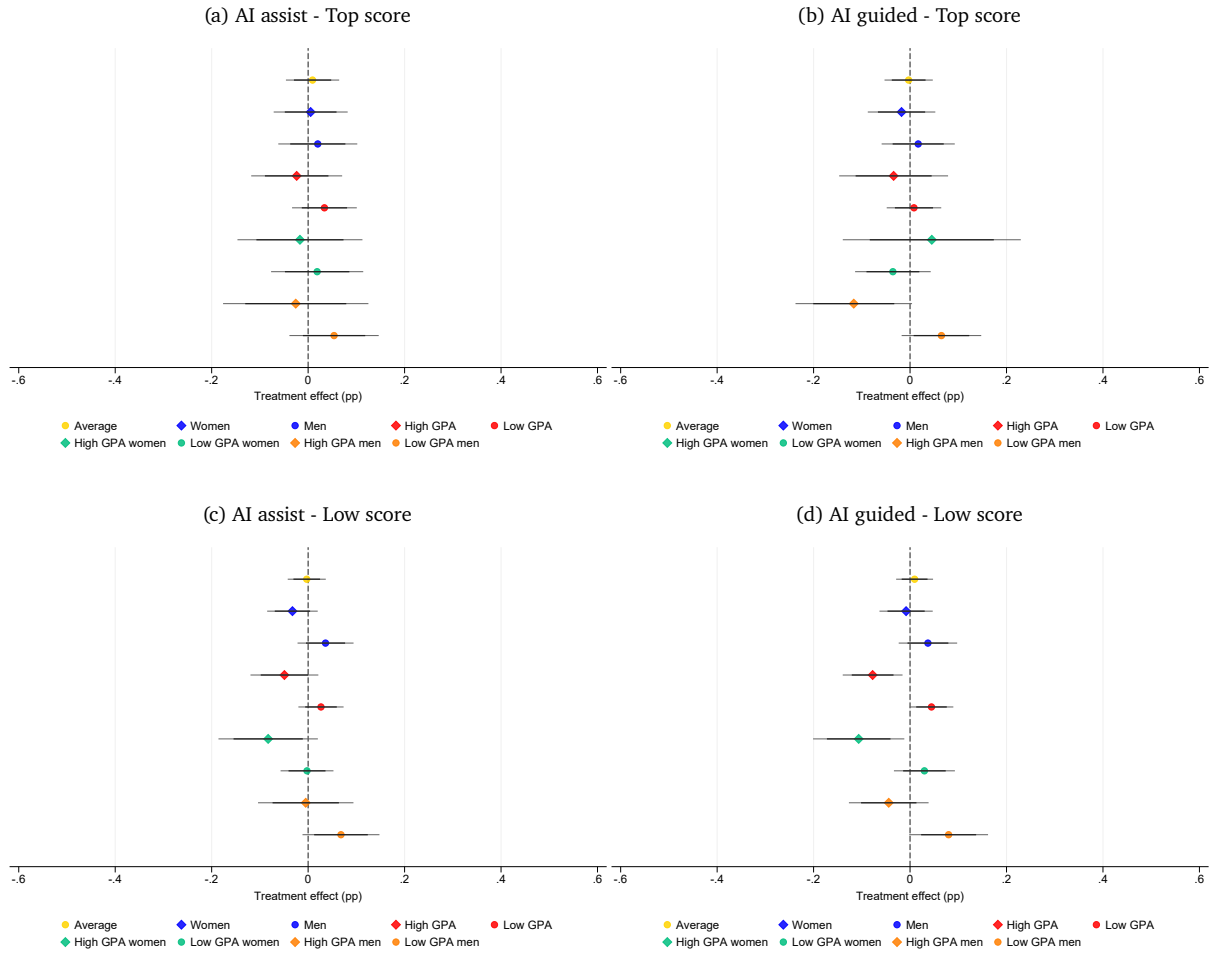
Notes. The plots report point estimates along with 95% confidence intervals of treatment effects on exam performance for the full sample (top coefficient) and for the subsamples listed below. 83% confidence intervals are added to facilitate visual comparisons of differences across subgroups. Sub-sample estimates are obtained by comparing treated students within a given subgroup to control-group students in the same subgroup. Panel (a) reports the effect of assignment to the AI-assisted treatment relative to the control group, and Panel (b) reports the effect of assignment to the AI-guided treatment relative to the control group. All regressions from with the coefficients are obtained control for: Age 21 or younger, undergraduate student, indicators for field of study, use AI occasionally or all the time, has a paid subscription to an AI provider. In regressions without interactions with gender and high GPA, we add these variables as covariates.

Figure 5: Treatment effects on self-assessed learning by subgroups



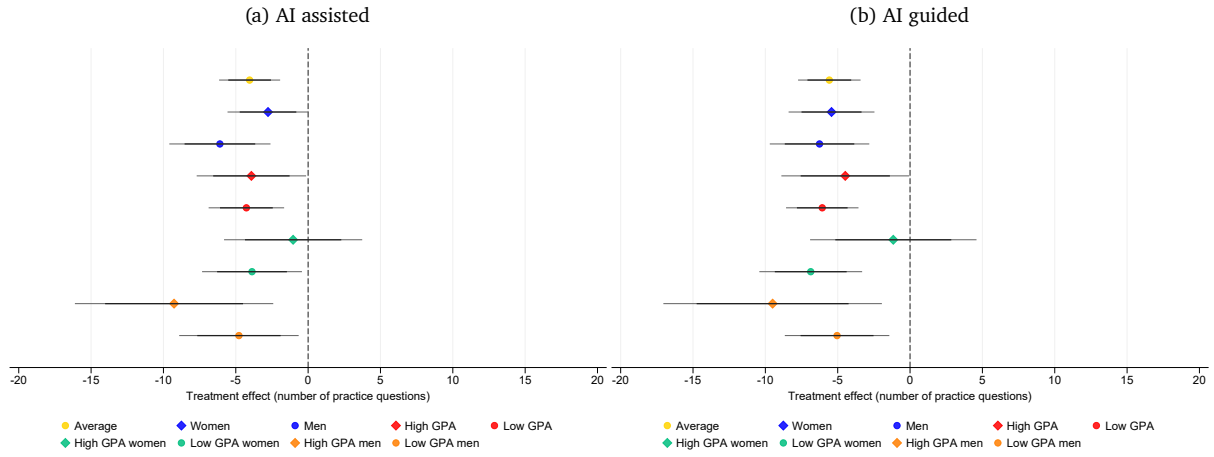
Notes. The plots report point estimates along with 95% confidence intervals of treatment effects on self assessed learning for the full sample (top coefficient) and for the subsamples listed below. 83% confidence intervals are added to facilitate visual comparisons of differences across subgroups. Subsample estimates are obtained by comparing treated students within a given subgroup to control-group students in the same subgroup. Panel (a) reports the effect of assignment to the AI-assisted treatment relative to the control group, and Panel (b) reports the effect of assignment to the AI-guided treatment relative to the control group. All regressions from with the coefficients are obtained control for: Age 21 or younger, undergraduate student, indicators for field of study, use AI occasionally or all the time, has a paid subscription to an AI provider. In regressions without interactions with gender and high GPA, we add these variables as covariates.

Figure 6: Treatment effects on scoring at the top or bottom



Notes. The plots report point estimates along with 95% confidence intervals of treatment effects on obtaining a top or low exam performance for the full sample (top coefficient) and for the subsamples listed below. 83% confidence intervals are added to facilitate visual comparisons of differences across subgroups. Subsample estimates are obtained by comparing treated students within a given subgroup to control-group students in the same subgroup. Panels (a) and (b) report the effect on obtaining a score higher than 10 in the exam, and Panels (c) and (d) report the effect on obtaining a score lower than 5. Both score thresholds were pre-registered. All regressions from with the coefficients are obtained control for: Age 21 or younger, undergraduate student, indicators for field of study, use AI occasionally or all the time, has a paid subscription to an AI provider. In regressions without interactions with gender and high GPA, we add these variables as covariates.

Figure 7: Treatment effects on number of practice questions attempted



Notes. The plots report point estimates along with 95% confidence intervals of treatment effects on number of practice questions attempted for the full sample (top coefficient) and for the subsamples listed below. 83% confidence intervals are added to facilitate visual comparisons of differences across subgroups. Subsample estimates are obtained by comparing treated students within a given subgroup to control-group students in the same subgroup. Panel (a) reports the effect of assignment to the AI-assisted treatment relative to the control group, and Panel (b) reports the effect of assignment to the AI-guided treatment relative to the control group. There were 90 practice questions in total. All regressions from with the coefficients are obtained control for: Age 21 or younger, undergraduate student, indicators for field of study, use AI occasionally or all the time, has a paid subscription to an AI provider. In regressions without interactions with gender and high GPA, we add these variables as covariates.

A Appendix Tables

Table A1: Share of correctly solved practice questions

	Success rate in attempted practice questions			
	(1)	(2)	(3)	(4)
AI-assisted	0.055*** (0.017)	0.016 (0.028)	0.038* (0.022)	-0.004 (0.035)
AI-guided	0.062*** (0.017)	0.027 (0.026)	0.056*** (0.020)	0.021 (0.031)
Female		-0.051** (0.023)		-0.048* (0.029)
AI-assisted × Female		0.063* (0.035)		0.064 (0.045)
AI-guided × Female		0.057 (0.035)		0.056 (0.041)
High GPA			0.008 (0.023)	0.008 (0.037)
AI-assisted × High GPA			0.042 (0.034)	0.044 (0.056)
AI-guided × High GPA			0.012 (0.036)	0.015 (0.057)
Female × High GPA				-0.006 (0.047)
AI-assisted × Female × High GPA				0.002 (0.070)
AI-guided × Female × High GPA				-0.000 (0.073)
Constant	0.685*** (0.028)	0.712*** (0.030)	0.692*** (0.028)	0.719*** (0.031)
Controls	Yes	Yes	Yes	Yes
Mean women	0.678	0.678	0.678	0.678
Mean men	0.736	0.736	0.736	0.736
Observations	595	585	595	585

Notes: The outcomes are at the top of the columns. Estimates based on the specifications and heterogeneity reported in our pre-analysis plan. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Phrasing of treatments

B.1 Control condition

You have been provided access to **Google Search and Google Translate** to assist you during the practice questions. These tools can help you look up information, find translations, and understand concepts from the material. You are free to decide how to use these tools to support your learning and answer the practice questions.

Here are some examples of how you might use the tools:

- You can find explanations or examples related to the grammar rules, such as:
 - “How do verbs change in Esperanto?” or “Esperanto plural nouns examples.”
- You can translate unfamiliar words or phrases between Esperanto and English.
- If you are curious about broader topics, you can search for additional context to enhance your understanding.

There are no specific instructions on how to use these tools, please use them as you normally would when you study for a topic in school. Some students may choose to use these tools frequently, while others may prefer relying on their notes or prior knowledge.

Reminders:

- The goal of this session is to engage with the material and complete the practice questions. Use the tools as you think is best to support your learning.
- You are encouraged to complete as many questions as possible within the time provided.
- There is no penalty for how much or how little you use the tools—you are free to decide how they fit into your learning process.

The goal is to learn as much as possible now. This is your chance to learn as much as possible with study aids to get the highest possible score in the final test where you will have no access to any external aids.

B.2 AI-assisted condition

You have been given access to a **premium ChatGPT** account to assist you during the practice questions. This tool can help you look up information, find translations, and understand concepts from the material. You are free to decide how to use this tool to support your learning and answer the practice questions. Other generative AI chatbots are disabled and you cannot use your own premium account if you have one.

Please go now to the ChatGPT premium account that is open in your computer. **Type the following: Date_time_computer number.** For example, if the date is 28th November, 2024, the time of your session is 15:00, and your computer is #11, please write on the ChatGPT input box: 28112024_1500_11 and hit enter. You will then write all your interactions with ChatGPT in the same screen.

Here are some examples of how you might use ChatGPT:

- You can find explanations or examples related to the grammar rules, such as:
 - *"How do verbs change in Esperanto?" or "Esperanto plural nouns examples."*
- You can translate unfamiliar words or phrases between Esperanto and English.
- If you are curious about broader topics, you can search for additional context to enhance your understanding.

There are no specific instructions on how to use this tool, please use it as you normally would when you study for a topic in school. Some students may choose to use this tool frequently, while others may prefer relying on their notes or prior knowledge.

Reminders:

- The goal of this session is to engage with the material and complete the practice questions. Use the tool as you think is best to support your learning.
- You are encouraged to complete as many questions as possible within the time provided.
- There is no penalty for how much or how little you use the tool—you are free to decide how they fit into your learning process.

The goal is to learn as much as possible now. This is your chance to learn as much as possible with study aids to get the highest possible score in the final test where you will have no access to any external aids.

B.3 AI-guided condition

You have been given access to a **premium ChatGPT** account to assist you during the practice questions. This tool can help you look up information, find translations, and understand concepts from the material. ChatGPT is a tool designed to help you with understanding the material and completing the practice questions. Other generative AI chatbots are disabled and you cannot use your own premium account if you have one.

Please go now to the ChatGPT premium account that is open in your computer. **Type the following: Date_time_computer number.** For example, if the date is 28th November, 2024, the time of your session is 15:00, and your computer is #11, please write on the ChatGPT input box: 28112024_1500_11 and hit enter. You will then write all your interactions with ChatGPT in the same screen.

To make the most of ChatGPT as a learning tool, follow these steps:

1. Explain your thought process or question. Before typing, **describe what you're trying to figure out or learn.** For example: *"I'm learning about Esperanto nouns and their endings. Can you explain how plural nouns work?"*
2. **Ask for step-by-step guidance.** Rather than asking for direct answers, request explanations or steps. For example: *"Can you explain step-by-step how to form a past-tense sentence in Esperanto?"*
3. Break questions into smaller parts. **Focus on one aspect at a time.** For instance:
"What is the ending for adjectives in Esperanto?"
Follow up with: *"Do adjectives also change in the plural form?"*
4. **Engage in follow-up questions.** Clarify and expand on ChatGPT's responses. For example: *"Can you give me more examples of how to use the word 'bela' (beautiful) in different contexts?"*
5. Practice actively. Write your own sentences and **ask ChatGPT to check them or provide feedback:**
"I wrote: 'La granda kato dormas.' Is this sentence correct?"

Reminders:

- The goal is to use ChatGPT not just to find answers but to deepen your understanding and learn how to approach the material thoughtfully.
- You are encouraged to ask questions, experiment with your understanding, and clarify concepts during the session.
- Take your time to engage fully with the tool and explore how it can help you learn.

The goal is to learn as much as possible now. This is your chance to learn as much as possible with study aids to get the highest possible score in the final test where you will have no access to any external aids.